mhs-poisson-z: pressure-only magnetohydrostatic correction (height grid)

Physics of `run.mode = "mhs-poisson-z"` (HSEMHS). 24 Oct 2026.

1. What the mode does and why

The full magnetohydrostatic solve (`full-mhs-z`) changes both the pressure and the magnetic field and needs a day on four nodes for the Bifrost cube. Inside a Stokes inversion, where the atmosphere changes at every iteration, that is far too slow. The FIRTEZ-dz inversion code (Borrero et al.) instead keeps T and $\mathbf{B}$ as given and corrects **only the pressure**, by solving a Poisson-type equation. This mode does the same on our grid with our equation of state: it starts from `column-hse-z`, keeps T and $\mathbf{B}$ fixed, and finds the pressure that balances gravity and the given Lorentz force as well as possible. It takes minutes.

2. Quantities

Symbol	Meaning	Unit
T, $\mathbf{B}$	temperature and magnetic field from the input, **both fixed**	K, T
$\mathbf{J}$, $\mathbf{L}$	current density $\nabla\times\mathbf{B}/\mu_0$; Lorentz force $\mathbf{L}=\mathbf{J}\times\mathbf{B}$ (fixed)	$\mathrm{A\,m^{-2}}$, $\mathrm{N\,m^{-3}}$
P, u	gas pressure, the only unknown; solved as $u=\ln P$	Pa, –
ρ	$\rho(T,P)$ from the EOS table	$\mathrm{kg\,m^{-3}}$
H, $\bar{H}$	scale height $P/(\rho g)$; its layer median in the starting state	m
β	plasma beta $2\mu_0 P/B^2$	–
z_{cut}	top of the solved region (production 1 Mm)	m

3. The equation

A static atmosphere has zero net force per unit volume,

$$\mathbf{F}\equiv-\nabla P+\mathbf{L}-\rho g\,\hat{\mathbf{z}}=0.$$

With $\mathbf{B}$ fixed, $\mathbf{L}$ is a given force field, and ρ depends on P only through the EOS. Taking the divergence,

$$\nabla^2 P=\nabla\cdot\mathbf{L}-g\,\partial_z\rho,$$

a Poisson equation for P; hence the name. In a horizontally periodic box it separates into Fourier modes, each a one-dimensional problem in z, which is what makes it fast.

4. What a pressure correction can and cannot do

Any force field splits uniquely into a **gradient part** $-\nabla\phi$ (curl-free) and a **non-gradient part S** (divergence-free, orthogonal to every gradient), with $\int|\mathbf{F}|^2=\int|\nabla\phi|^2+\int|\mathbf{S}|^2$. A pressure change adds a pure gradient, $-\nabla\delta P$. If ρ were fixed, it could remove the gradient part exactly and never touch $\mathbf{S}$.

Which forces are non-gradient? The curl of the force balance is

$$\nabla\times\mathbf{F}=\nabla\times\mathbf{L}-g\,(\partial_y\rho,\,-\partial_x\rho,\,0),$$

so static balance needs the **curl condition** $\nabla\times(\mathbf{J}\times\mathbf{B})=g(\partial_y\rho,-\partial_x\rho,0)$: the magnetic tension must hold up every horizontal density contrast. Gravity is a gradient only if ρ is horizontally uniform; buoyancy of hot granules next to cool lanes is non-gradient. Without a field the only static atmosphere is horizontally uniform. With Bifrost's T and fixed $\mathbf{B}$ the curl condition does not hold, so *no pressure can make the atmosphere exactly static.*

The density loophole. ρ is not fixed: with T fixed, $\delta\rho=\rho\,(\partial\ln\rho/\partial\ln P)_T\,\delta P/P$, and $\delta(\nabla\times\mathbf{F})=-g\nabla\times(\delta\rho\,\hat{\mathbf{z}})$. By changing the pressure, and hence the density, the correction can shrink horizontal density contrasts and part of $\mathbf{S}$, but not the part due to the temperature contrast at fixed pressure or to the tension of $\mathbf{B}$. Measured on the production cube (original script, 23 Oct): column HSE is 80–90 % gradient error in the photosphere; the Poisson correction cut the gradient part 3.5–5 times and, through the density, the non-gradient part by 1.4–3.2 between 0 and 0.75 Mm.

5. The quantity minimised

The force is divided by P and the unknown is $u=\ln P$:

$$\mathbf{R}\equiv\frac{\mathbf{F}}{P}=-\nabla u-\frac{\hat{\mathbf{z}}}{H(T,P)}+\frac{\mathbf{L}}{P},\qquad \Phi(u)=\sum_{\text{centres}} w\,\bar{H}^2\big(R_x^2+R_y^2\big)+\sum_{z\text{-faces}} h\,\bar{H}^2R_z^2.$$

- $P = e^u$ stays positive across the many decades in the box.
- $\mathbf{R}$ is a *relative* imbalance: minimising $|\mathbf{F}|^2$ itself rewards lowering P (the first version drove P toward zero in 637 000 cells).
- $\mathbf{R}\bar{H} \approx \mathbf{F}/(\rho g)$: the imbalance is measured in units of the weight of the gas, layer by layer, so the dense deep layers do not swamp the upper ones. w, h are the cell heights.
- Without $\mathbf{L}$, $R_z = 0$ is $du/dz = -1/H$, i.e. column HSE.

The mean of $\ln P$ on the bottom plane is held at its column-HSE value: the overall pressure level is set by the input, not by the correction.

6. The low-β problem and $z_{\rm cut}$

Where $\beta \ll 1$ the term $\mathbf{L}/P$ dominates. Static balance there requires a force-free field, which a snapshot of a moving simulation is not; with $\mathbf{B}$ fixed, the only way to balance a large Lorentz force is a large pressure gradient. Because $\partial P/\partial z$ couples the layers, this inflates the pressure in the photosphere too: solved up to 3 Mm the median $\ln P$ rose by 3.2 (a factor 25); with the Lorentz force switched off nothing happened. The force balance is therefore solved only below $z_{\rm cut} = 1$ Mm (median $\beta \approx 8$); above it each column is integrated hydrostatically from the solved pressure at $z_{\rm cut}$. FIRTEZ makes the same choice in a softer way (`LIMIT_MAGTERM`). Above $z_{\rm cut}$ the atmosphere is hydrostatic, **not** magnetohydrostatic.

7. Optical depth

As in `column-hse-z`: τ_{500} is integrated down from the top of the box with the `exp_tail` closure, and $z(\tau_{500} = 1)$ is reported per column.

8. Results on the production cube

Mode run of 24 Oct (8 ranks on 4 nodes, 4.7 min, column HSE included): $\ln(P/P_{\rm colHSE})$ below $z_{\rm cut}$ has percentiles $1/50/99 = -0.54/+0.06/+0.99$; $z(\tau_{500} = 1)$ percentiles $1/50/99 = -5/69/125$ km against $-29/64/127$ km in column HSE. The objective fell to 8 % of its starting value; the run used the then step limit (60) while still improving by 2×10^{-4} per step. Force imbalance per layer from the original script (23 Oct, same physics), RMS of $|\mathbf{F}|$ over the layer median of ρg:

z [Mm]	column HSE	Poisson correction	Bifrost snapshot	`full-mhs-z` (M5h, iteration 132)
−0.5	0.11	**0.06**	0.11	0.09
0.0	0.33	**0.13**	0.29	0.27
0.25	0.23	**0.07**	0.12	0.19
0.5	0.22	**0.06**	0.12	0.18
0.75	0.28	**0.09**	0.21	0.22
1.0	0.66	**0.34**	0.49	0.51

This diagnostic uses the same discretisation the Poisson step minimises, which favours it; `full-mhs-z` balances forces on its own staggered grid.

9. What it means

- Where photospheric lines form ($z \lesssim 0.5$ Mm) the correction removes most of the force imbalance of column HSE, below that of the Bifrost snapshot itself, and barely moves the τ scale.
- The remaining imbalance is the buoyancy of the fixed temperature structure that the fixed field cannot hold up. Removing it needs $\mathbf{B}$ (or T) to change: `full-mhs-z`.
- $z_{\rm cut}$ is a physical choice: where the fixed field stops being trusted to be in equilibrium.